

Patho Study Guides 1-5 (filled)

NRS224 Pathophysiology · Exam 1 covers Classes 1-5 (Tue Oct 13)

Personal study sheets — not for submission.

- **1** · Cell, part 1 — filled
- **2** · Cell, part 2 (replication, cancer, genetics) — filled
- **3** · Hematopoietic function (Blood) — *class study guide not found on the box*
- **4** · Immune, part 1 — *class study guide not found on the box*
- **5** · Immune, part 2 (autoimmunity, inflammation, hypersensitivity) — filled

Guides 3 and 4 were not in any downloaded folder, so they are not filled here. Your full study sheet already has Blood (pp. 8-13) and Immune 1 (pp. 14-16).

Class 1 — Alterations in Cell Function & Growth, Part 1 (The Cell, part 1)

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 1-4.

Page 1 — The cell: components & functions

Guide prompt: "If pathological changes occur in any of these components → Nursing problems:"

1.1 → problems with nucleus:

- Nucleus = control center → DNA holds instructions for protein synthesis; RNA carries them to cytoplasm [S3-S4, 02:54-06:32; sheet p.1]
- Damaged nucleus → instructions get messed up → can't make certain proteins at all, OR proteins made but defective → don't work [S4, 04:08-06:32]
- Bottom line: trouble with protein synthesis [S4, 04:08-06:32; sheet p.1]

1.2 → problems with ribosomes:

- Ribosomes make proteins (site of protein synthesis out in cytoplasm)

[S6, 07:01-07:34; sheet p.1]

- Damaged ribosomes → same problem: proteins not made, or made defective and won't work [S6, 07:01-07:34]
- Class practice Q: "If the function of ribosomes were impaired..." → best answer: "Synthesis of defective proteins, or decreased protein synthesis" [class slide 6]

1.3 → problems with ER:

- ER = system of tubules: (1) transports substances from place to place inside cell, (2) houses ribosomes + enzyme systems (enzymes = catalysts that speed up chemical reactions) [S8, 07:54-09:42; sheet p.1]
- Problems: can't get things place to place; ribosomes have nowhere to hang out → protein synthesis problems; chemical reactions slow way down [S8, 07:54-09:42]
- Class practice Q: impaired ER → best answer: "Substances may not be transported appropriately within the cell" [class slide 7]

1.4 → problems with mitochondria:

- Mitochondria = powerhouse ("Mighty Chondrion") → transforms food energy into ATP; needs oxygen to do it [S11, 10:28-12:06; sheet p.1]
- Not enough O₂ → can't make ATP → power failure → cell can't carry out activities → dies [S11, 10:28-12:06]
- Her examples: heart attack = heart muscle cells die from power failure; stroke = brain cells [S11, 10:28-12:06]

1.5 → problems with lysosomes (and peroxisomes):

- Lysosomes = bags of digestive enzymes → get rid of worn-out cell parts / foreign material (e.g., bacteria) [S12, 12:06–12:33; sheet p.1]
- Not working → things accumulate inside cell = lysosomal storage disease → cell damaged or malfunctions [S13–S14, 12:33–13:08]
- Example: Tay-Sachs (missing lysosomal enzyme → accumulation in retina + nervous system → blindness, motor/mental issues) [S15, 13:08–14:15]
- Peroxisomes = specialized lysosomes that detoxify/get rid of free radicals (unstable atom/molecule that can damage cells) → free radical damage blamed for wrinkles, cancer, atherosclerosis, heart disease [S19, 15:16–16:36; sheet p.1]
- ★ **Prof emphasis:** she won't test on Tay-Sachs specifics — "more likely to ask" what happens if lysosomes aren't working [S18, 14:43–15:16]

1.6 → problems with Golgi complex:

- Golgi = post office of the cell → takes ribosome-made proteins, packages them into secretory vesicles/granules, directs them to the right place [S20–S21, 16:36–17:55; sheet p.1]
- Not working → proteins made don't get to the right place → can't do their job [S21, 17:06–17:55]
- Example: I-cell disease — lysosome enzyme packaged/released into blood → lysosome missing enzyme (storage disease) + enzyme

breaks down things in blood you still want → damage to heart, lungs, nervous system, MSK deformities [S22-S23, 17:55-20:36]

- ★ **Prof emphasis:** no test questions specifically on I-cell disease — only the underlying pathology of lysosome/Golgi malfunction [S23, 19:23-20:36]

1.7 → problems with cell membrane:

- Membrane holds cell together — keeps what belongs inside in, keeps out what doesn't; carrier/transport proteins + ion channels let O₂/fuel in and waste out [S24, 20:36-21:35; sheet p.1]
- Receptor sites (hormones/chemical messengers) → cellular communication; damaged receptors → poor communication AND medications that stimulate/block receptors won't be effective [S26, 22:11-24:26]
- Self-antigens tell immune system "self vs non-self" → if damaged, immune system may attack self-cells = autoimmune disease [S26, 22:11-24:26]
- Also maintains resting membrane potential + generates action potentials; worst case: cell breaks apart and dies [S26, 22:11-24:26]

Page 2 — Cell metabolism & movement across the membrane

2.1 Nursing problem (altered cellular metabolism/energy production):

- Cellular metabolism = cell makes energy: food → ATP, stored until needed [S27, 24:26-25:26]

- Two pathways: aerobic (requires O_2 , done in mitochondria) vs anaerobic (no O_2 , less efficient, fewer ATP, but a good backup)
[S27-S29, 24:26-26:26; sheet p.2]
- Anaerobic too long → lactic acid → changes body fluid pH = acidosis → lots of things don't function normally [S30, 26:26-27:53]
- 3 things cell needs for energy: fuel, oxygen, appropriate enzymes (catalysts) → missing any one → not enough energy → cells malfunction or die [S30, 26:26-27:53; sheet p.2]

2.2 Pathological change in movement across membrane → Nursing problem:

- 3 ways across membrane: passive movement, active transport, endocytosis/exocytosis
[S31, 27:53-29:00; sheet p.2]
- Passive = down concentration gradient (high → low), no energy, goal = equilibrium; diffusion = particles move, osmosis = water moves; deciding factor = membrane permeability [S32-S43, 29:00-34:32]
- Hypernatremia example: too much Na^+ in blood → bigger gradient → Na^+ still can't cross → MORE water osmoses out → cell shrinks → brain cells shrink = lots of neuro issues
[S44-S47, 34:32-37:04]
- Active transport = against gradient, requires energy (ATP) — Na^+/K^+ pump keeps K^+ in, Na^+ out → needed for action potentials
[S48-S49, 37:04-40:38]
- Endocytosis (pinocytosis = fluid/dissolved "cell drinking"; phagocytosis = solids, e.g., neutrophils/monocyte-macrophages eating

bacteria) → problem = big impact on immunity; exocytosis releases hormones/gets rid of waste → problem = can't release hormones [S50, 40:38-42:58]

Page 3 — Cellular communication & membrane potentials

3.1 Nursing problem (alterations in availability/function of chemical messengers or receptor sites):

- Chemical messenger method: one cell releases hormone/neurotransmitter/growth factor → binds receptor site on 2nd cell → tells that cell what to do [S51, 42:58-46:13; sheet p.2]
- Two places it breaks: (1) too much or too little messenger, (2) damaged or missing receptor sites → message never gets delivered [S51, 42:58-46:13]
- Her examples: Parkinson's = not enough dopamine (tremor, slow stiff movements); schizophrenia = too much dopamine; myasthenia gravis = autoimmune attack on receptors at neuromuscular junction → muscle weakness [S51-S52, 42:58-47:34]
- Predisposing factors per sheet: autoimmune dz, congenital defects, genetic predisposition, degeneration d/t age or toxin exposure [sheet p.2]
- Meds work by increasing/decreasing messenger or stimulating/blocking receptors [S52, 46:13-47:34; sheet p.2]

3.2 Nursing problem (changes in membrane permeability):

- Resting membrane potential: cell electrically polarized, inside more negative (K^+ diffuses out, Na^+ can't get in → mostly anions left inside) [S54–S58, 49:37–52:59; sheet p.3]
- Action potential: something changes membrane permeability → Na^+ diffuses in → inside becomes positive = depolarization; ions then pass cell-to-cell through gap junctions (nerve/muscle communication) [S59–S61, 52:59–55:25]
- Repolarize (2 steps): membrane goes back to not letting Na^+ in + Na^+/K^+ pump pumps K^+ back in, Na^+ back out → ready to repeat [S62–S64, 55:25–57:00]
- Problems: electrolyte imbalance messes up the gradient; hypoxia/impaired metabolism (no O_2 or glucose) → no ATP → no Na^+/K^+ pump → can't repolarize → brain & heart can't function [S66, 57:19–59:30; sheet p.3]

Page 4 — Cell injury: adaptive responses, reversible injury, cell death

Guide lists each response with space under it. Fill: what it means clinically, upside/downside, and her example(s).

4.1 atrophy

- Cells get smaller / revert to lower level of functioning when not enough O_2 , fuel, or blood flow → smaller cell needs less O_2 /nutrients → more likely to survive [S79, 74:33–76:03; sheet p.4]

- Upside: cell survives, doesn't die. Downside: not as functional [S79, 74:33-76:03]
- Her examples: calf muscle atrophy from chronic impaired arterial blood flow → can't do as much work; atrophied endocrine gland → less hormone [S80, 76:03-76:47]

4.2 hypertrophy

- Cells get bigger in response to increased work demands — opposite of atrophy [S81, 76:47-77:05; sheet p.4]
- Examples: bodybuilders lifting heavier weights (skeletal muscle cells); left ventricle in chronic high BP (pushing against high arterial pressure → heart muscle cells hypertrophy) [S82-S83, 77:05-77:56]
- Upside: cells can do more work. Downside: bigger cells doing more work need more blood, O₂, fuel → if atherosclerosis limits supply → may set person up for a heart attack [S84, 77:56-78:40]

4.3 hyperplasia

- Increase in cell NUMBER (not size) — usually in response to chemical stimulus: change in hormone or growth factor [S85, 78:40-80:31; sheet p.4]
- Examples: wound healing (growth factor → more cells to fill in wound); pregnancy uterus = BOTH hypertrophy + hyperplasia [S85, 78:40-80:31]
- Downside: tissue enlarges enough to interfere with surrounding structures — benign

prostatic hyperplasia squeezes urethra shut → trouble emptying bladder [S86, 80:31-81:29]

- ★ **Prof emphasis:** "students do mix those two terms up a lot on the test" — hypertrophy = size, hyperplasia = number [S85, 78:40-80:31]

4.4 metaplasia

- Cells change from one cell type to another that's better able to survive — triggered by chronic irritation/inflammation [S87, 81:29-82:19; sheet p.4]
- Not a change out of primary type (not nerve→muscle); e.g., one kind of epithelial cell → a sturdier one [S87, 81:29-82:19]
- Her example: smoker's airway ciliated columnar epithelium → stratified squamous (sturdier, less likely to be injured) [S88, 82:19-83:14]
- Downside: lose protective cilia → mucus + trapped bacteria/irritants get stuck in airway, can't get out [S88, 82:19-83:14]

4.5 dysplasia

- Book's definition she likes: "deranged cell growth" — cells can be all different shapes and sizes, don't really look like each other [S90, 83:45-84:27; sheet p.4]
- Usually a reaction to chronic irritation/inflammation; progression: normal → metaplasia → dysplasia if irritation isn't removed [S90-S91, 83:45-86:00]
- No upside; big downside = precursor to cancer (higher risk, NOT a guarantee) [S91, 84:27-86:00]

- Examples: airways of smokers; uterine cervix — Pap smear finding dysplastic cells $\neq$ cancer, means higher risk $\rightarrow$ more frequent Paps

[S91, 84:27-86:00]

4.6 reversible injury

- Cellular response ladder depends on severity of stressor: adapt $\rightarrow$ become injured $\rightarrow$ die
[S78, 74:10-74:33; sheet p.3]
- If the stress is severe the cell becomes injured anyway; if we get rid of the source of the stress, cell may heal and injury be reversed
[S92, 86:00-89:15]
- If injury is bad enough, the cell will actually die ($\rightarrow$ 4.7) [S92, 86:00-89:15]

4.7 cell death/necrosis - if cell injury is irreversible

- How bad depends on how many cells die: a few liver cells = probably fine; a whole bunch = liver failure $\rightarrow$ patient may die
[S92, 86:00-89:15]
- Replacement varies: skin cells replace themselves all the time; brain cells do so at a much, much slower rate [S92, 86:00-89:15]
- Dead cells replaced either by same cell type OR by scar tissue — heart attack: heart muscle cells replaced with scar tissue $\rightarrow$ restores structure but NOT function (no action potentials, no contraction) $\rightarrow$ chronic heart failure [S92, 86:00-89:15]
- Lab markers of cell damage: draw blood, look for proteins/enzymes normally found only inside cells $\rightarrow$ high levels = cells damaged,

insides leaking out. Troponin → heart muscle cells (chest pain in ED) [S74, 69:15-70:51]

- CK and LD → "damage to the liver or the skeletal muscles or some tissues like that" — she does not pair them one-to-one (your sheet writes CK → skeletal muscle, LD → liver)

[S74, 69:15-70:51; sheet p.3]

Class 2 — Alterations in Cell Function & Growth, Part 2 (Cell replication, cancer & genetics)

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture audio, transcribed by machine (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 5-7.

Page 1 — Cell replication/differentiation → Cancer

Guide: Mutations in regulatory genes → abnormal proliferation + abnormal differentiation. Example of disorder: cancer. Each printed finding has space under it; fill with what it means / what you'd assess.

1.1 → uncontrolled cell growth

- More cells actively cycling; fewer in the resting phase → uncontrolled growth.
[S10, 08:59-12:15]
- Lose cell density-dependent inhibition; do not die within their normal programmed time frame. [S20, 28:54-30:09]
- Increasing tumor cells crowd out normal cells and steal their blood supply.
[S21, 30:09-30:33; sheet p.6]

1.2 → infiltration into surrounding tissues

- Infiltration = growth extending into surrounding normal tissue; malignant tumors lack a containing capsule. [S38, 50:34-51:10]
- Loss of cell-to-cell cohesiveness → cells break away and travel to other body sites: metastasis. [S37, 47:31-50:34]
- Infiltration makes complete surgical removal difficult; remaining cancer cells keep multiplying → recurrence.
[S37, 47:31-50:34; S38, 50:34-51:10]

1.3 → tissue/organ dysfunction

- Abnormal differentiation → cancer cells do not specialize properly or perform normal tissue functions. [S22, 30:33-31:33]
- Normal cells are crowded out while abnormal cells do not function normally → tissue/organ dysfunction. [S21, 30:09-30:33; S22, 30:33-31:33]

1.4 → changes in cell characteristics

- Cells fail to differentiate/specialize properly → different characteristics and impaired normal function. [S22, 30:33-31:33; S23, 31:33-32:12]
- Lose normal cell-to-cell cohesiveness → break away and metastasize. [S37, 47:31-50:34]
- May produce abnormal substances and display abnormal cell-membrane antigens.
[S24, 32:12-32:31; S29, 36:43-37:09]

1.5 abnormal biochemical products (ex: VEGF, ectopic hormone production, toxins, etc.)

- VEGF = vascular endothelial cell growth factor
→ new blood vessels supply oxygen/nutrients, allowing continued tumor growth.

[sheet p.6; S25, 32:31–33:37]

- Ectopic hormones: cancer cells make hormones without normal negative-feedback regulation → excess hormone/endocrine-like disease. [S26, 33:37–35:49]
- Enzymes/toxins damage tissue → sores that do not heal normally. [S27, 35:49–36:32]
- Abnormal antigens such as PSA and CEA can help monitor cancer progression and treatment response; professor describes this role rather than diagnosis/screening.

[S29, 36:43–37:09; S30, 37:09–40:49]

1.6 abnormal antigens

- Abnormal differentiation → abnormal antigens on cancer-cell membranes; these can serve as clinical markers. [S29, 36:43–37:09]
- Immune system normally fights cancer; immune defects/immunosuppression increase cancer risk. [S18, 25:49–27:10; S44, 58:23–61:14]
- Examples of increased risk: AIDS, immunosuppressive drugs, and aging with less efficient immune function. [S18, 25:49–27:10]

1.7 generalized effects

- Cachexia = weight loss; WBC cytokines cause anorexia/loss of appetite → reduced eating

and weight loss. [S33, 42:10–42:53; S34, 42:53–43:48]

- Pain: tumor pressure on nerves, inadequate blood/oxygen, and tissue damage from enzymes/toxins. [sheet p.6; S35, 43:48–45:34]
- Anemia: poor nutrition (iron/B-vitamin deficiency), bleeding from damaged tissue, or tumor erosion into vessels. [S36, 45:34–47:31]
- Bone-marrow involvement impairs RBC production; cancer treatments can also cause anemia. [S36, 45:34–47:31]

1.8 Example of disorder: benign neoplasms — Nursing problems:

- Benign = noncancerous: abnormal proliferation only, with normal differentiation; malignant = both abnormal. [S37, 47:31–50:34]
- Usually fibrous capsule → contained expansion, no infiltration or metastasis; easier to remove surgically. [S39, 51:10–51:43; sheet p.6]
- Nursing problems are mainly local effects of increasing size/location—not the abnormal substances made by cancer cells.
[S40, 51:43–53:39]
- Can still be fatal if growth interferes with a vital function; benign brain tumors are an example. [S40, 51:43–53:39]

1.9 Interventions (cancer / neoplasms) — blank Interventions column

- Surgery: remove tumor; infiltrating cancer is harder to remove completely, and remaining cells can regrow. [S37, 47:31–50:34; S41, 53:39–54:47]
- Radiation: generates free radicals that damage cancer cells/interfere with replication;

rapidly proliferating cells are most affected, but normal cells can be injured.

[S42, 54:47-55:56; class slide 20]

- Chemotherapy interferes with the cell cycle: S-phase drugs inhibit DNA replication; M-phase drugs prevent division. [S10, 08:59-12:15]
- Different phase targets explain combination drugs and treatment long enough for cells to reach the targeted phase. [S10, 08:59-12:15]
- Chemo effects are not confined to cancer → side effects; cancer treatment may cause anemia. [S43, 55:56-58:23; S36, 45:34-47:31]
- Biologic response modifiers/immunomodulators modify immune responses: monoclonal antibodies and synthetic cytokines. [S44, 58:23-61:14]

1.10 Pathological change: Mutations in regulatory genes — which genes? (proto-oncogenes/oncogenes, tumor-suppressor genes) and PF details

- Genetic mutations alter programming that controls proliferation and differentiation; specific proto-oncogenes/oncogenes and tumor-suppressor genes are not named in these sources. [S13, 15:56-18:11; S14, 18:11-20:45]
- Carcinogens: chemicals, radiation (including sun), cigarette exposure; free-radical DNA damage can cause mutations. Exposure-to-diagnosis lag may be 5-30 years.

[S15, 20:45-23:10; sheet p.6]

- Cancer-causing viruses listed: HPV, EBV, HBV; incorporation of viral DNA can alter host-cell

instructions → abnormal replication.

[sheet p.6; S17, 24:08–25:49]

- Immune defects: AIDS, immunosuppressive drugs, aging → increased cancer incidence. [S18, 25:49–27:10]
- Heredity: germ-cell mutations can pass to children; mutations in regular body cells do not. Cancer risk usually reflects a combination of factors. [S19, 27:10–28:54]

Page 2 — Genetic aspects of the cell

2.1 Gene mutation → Nursing problem:

- DNA replication error/mutation → altered instructions for a protein; radiation or toxic exposure may contribute. [S10, 08:59–12:15; S61, 78:08–79:26]
- Protein is absent or defective/nonfunctional → nursing problems depend on that protein's role. [S62, 79:26–82:02]
- Severity ranges from mild to fatal; no single set of findings applies to all single-gene disorders. [S62, 79:26–82:02]
- Germ-cell mutation can be passed to the next generation. [S61, 78:08–79:26]

2.2 Sex-linked: hemophilia A (factor VIII) — nursing problems/findings

- Hemophilia A: deficiency/nonfunction of factor VIII → assess muscle hematomas and hemarthrosis (joint bleeding). [sheet p.7]
- X-linked recessive pattern: males with a defective X usually have disease; females with

one defective X are usually carriers.

[S66, 84:28-85:47; S67, 85:47-87:03]

- ★ **Prof emphasis:** sex-linked test questions specify a son or daughter—calculate risk using the two possibilities of that sex, not all four children. [S84, 98:48-100:21]

2.3 Autosomal dominant: Marfan's syndrome — nursing problems/findings

- Autosomal dominant: one defective gene causes disease; Aa = disease, aa = normal; no carriers in the taught pattern.
[S72, 88:29-89:57]
- Defective connective tissue → long, thin arms and fingers; musculoskeletal deformities.
[S73, 89:57-90:47; S75, 91:09-91:49]
- Assess eye problems: retinal detachment or lens dislocation from weak connective tissue/ligaments. [S75, 91:09-91:49]
- Assess cardiovascular problems: valve prolapse and aneurysms. [S75, 91:09-91:49]

2.4 Autosomal recessive: sickle cell anemia — nursing problems/findings

- Autosomal recessive: two defective genes needed; aa = disease, Aa = carrier, AA = normal genes. [S76, 91:49-93:44]
- Abnormal RBCs clump together → obstruction/clot-related problems.
[S77, 93:44-95:07]
- Hemolytic component: breakdown of RBCs/hemoglobin contributes to anemia.
[S78, 95:07-95:58]

2.5 Chromosomal disorders: Trisomy 21 (Down syndrome) — findings; also mitosis vs meiosis errors (nondisjunction)

- Mitosis: somatic/body cells; meiosis: germ cells, with 23 chromosomes per egg/sperm → 46 at fertilization. [S4, 01:32-02:10; S9, 07:55-08:59]
- DNA-copying errors can cause gene disorders; chromosome-separation errors cause unequal chromosome numbers. [S10, 08:59-12:15]
- Meiotic nondisjunction: one daughter cell receives both copies, the other none → after fertilization, trisomy or monosomy.
[S95, 117:31-118:28; S96, 118:28-119:10; S98, 120:06-120:38]
- Down syndrome = trisomy 21: three copies of chromosome 21. [S99, 120:38-122:13]
- Findings: characteristic facial features, intellectual disability, possible congenital heart problems and GI malformations.
[S99, 120:38-122:13]
- Structural errors can also occur: a chromosome piece breaks off/is lost → loss of multiple genes. [S93, 114:36-116:33]

2.6 Interventions (genetic disorders) — blank Interventions column

- Single-gene interventions depend on the affected protein and problems caused; no universal treatment. [S62, 79:26-82:02]
- Chromosomal disorders: treat symptoms, maximize quality of life, and help families support development. [S100, 122:13-124:03]

- Genetic counseling: Punnett squares estimate the chance of an affected child, carrier, or child with normal genes. [S80, 96:17–97:13]
- ★ **Prof emphasis:** Punnett squares will be on the test—diagram parents, fill boxes, label each child's status, then calculate the specific risk asked. [S81, 97:13–97:41]
- For her sex-specific X-linked questions, use X/Y and risk out of 2; for autosomal questions, use A/a and risk out of 4.

[S84, 98:48–100:21; S88, 106:29–108:06; S90, 110:58–112:26]

Class 5 — Immunity & Inflammation, Part 2: Autoimmunity & Hypersensitivity

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture subtitles (times are the 1x recording, about ± 30 s; on the 2x video, halve them), recorded and class slides, verified clinical excerpts, and your full study sheet pp. 17–19.

Page 1 — Distinguishing self from nonself

Guide: "Self-tolerance (the absence of an immune response directed against a person's own antigens) normally is maintained through:" — two blank dashes, then MHC molecules (3rd mechanism, already printed).

A1 Self-tolerance mechanism #1 (blank -)

- Mechanism #1 = regulatory T cells (aka suppressor T cells) [02:13]
- Two jobs: shut off immune response once pathogen is gone; suppress any potentially self-reactive WBCs [02:13]
- Gets rid of those self-reactive cells before they can engage in autoimmune activity [02:48]
- Sheet version: turn off immune response + suppress self-reactive WBCs [sheet p.17]

A2 Self-tolerance mechanism #2 (blank -)

- Mechanism #2 = bone marrow + thymus screening of immature WBCs [02:48]
- T cells mature in thymus, B cells mature in bone marrow; both sites screen for potentially self-reactive cells [02:48]
- If found → get rid of self-reactive cells BEFORE release into general circulation [03:20]
- Sheet version: screen + remove immature self-reactive B cells and T cells [sheet p.17]

Page 1 — Inflammation

B1 S/S (cardinal signs): list all 5 (fills the five blank dashes) and for each, the mechanism she gave

- Redness + heat ← vasodilation → more blood to area [58:27]
- Swelling (edema) ← ↑ capillary permeability → fluid leaks into tissues [59:31]
- Pain ← swelling; usually goes along with it [59:31]
- Impaired/loss of function ← swollen parts don't work (e.g., sprained finger can't bend) [1:00:02]
- Cardinal signs listed: redness, heat, swelling, pain, impaired function [sheet p.18]

B2 Vascular response: momentary vasoconstriction, then vasodilation → ____

- Momentary vasoconstriction → vasodilation [sheet p.18]
- Vasodilation = first vascular event → brings more blood: O₂, fuel, antibodies to fight

infection [58:27]

- Result = the redness and warmth of inflammation [58:59]

B3 ↑ capillary permeability → fluid escapes into tissues → ____

- ↑ capillary permeability = capillaries get 'leaky' → fluid leaks into tissues, bringing nutrients + antibodies [59:31]
- Causes edema, pain, and loss of function [59:31]
- Exudate types: serous (clear, light yellow = serum), serosanguineous (watery blood/pink), purulent (pus = dead WBCs, infected) [1:01:08]; [slide 53]
- Exudate is usually GOOD: brings nutrients/O₂/antibodies, flushes dirt, dilutes irritants, slows spread of contaminants [1:02:13]

B4 Cellular response - movement of WBCs into area of injury → ____

- Cellular response = phagocytic WBCs migrate to the injury area [1:03:53]
- Neutrophils get there first, followed by monocyte-macrophages [1:03:53]
- They phagocytize pathogens + dead tissue → get it ready for repair [1:03:53]
- If needed, they act as APCs → signal/activate adaptive immunity [1:04:24]

B5 Acute-phase response (fever, ↑ # WBCs, skeletal muscle catabolism) — what to know

- Acute-phase response = systemic effects: fever, leukocytosis, ↑ inflammatory markers, varying degrees of skeletal muscle catabolism [slide 57]; [1:05:39]
- Fever is beneficial: bacteria can't survive higher temp + WBCs become more active [1:05:26]
- Leukocytosis = actual ↑ in number of WBCs; fever + ↑ WBC ≠ always infection (post-op incision, MI) → monitor trends, check pus, wound culture [1:05:39]
- Skeletal muscle catabolism = break down muscle protein → amino acids to make antibodies / repair tissue; body mobilizes only what's useful (mosquito bite vs appendicitis) [1:10:21]
- Markers: CRP (liver; regulates cytokines), fibrinogen (helps locate pathogens), ESR (markers stick to RBCs → heavier → sediment faster) [36:36]; [37:37]
- ★ **Prof emphasis:** 'those are the only three I'm going to expect you to recognize for the test' (CRP, fibrinogen, ESR) [39:49]

B6 Potential problems r/t inflammation

- Inflammation can get out of control: very severe, or recurrent/persistent → harmful [1:12:01]
- Chronic irritation/inflammation → metaplasia + scar tissue formation [1:12:34]

- Scar tissue → loss of function + deformity (RA joints pulled into abnormal positions) [1:13:09]
- Kidney scarring → can't filter blood / make urine like real kidney tissue [1:13:41]

B7 Potential problems if immune system function impaired/altered

- Immunodeficiency → infection [class slide 4]
- Hypersensitivity → allergic reactions → unnecessary inflammation [class slide 4]
- Autoimmunity → damage to self tissue [class slide 4]
- Diagnostics: immunodeficiency → cultures, WBC count; hypersensitivity → allergy testing; autoimmunity → check for autoantibodies [class slide 5]
- ★ **Prof emphasis:** ALL three can cause inflammatory markers to increase [class slide 5]

Flowchart 1 — Autoimmunity (breakdown in self-tolerance)

C1 Nursing problem:

- Nursing problem: damage to self tissue — immune system attacks self-antigens [class slide 4]
- Injured self cells → organ/body system not working properly [22:17]
- Most autoimmune diseases are chronic: cycles of remissions and exacerbations/flare-ups [32:40]
- Goal of care: minimize exacerbations, get patients into remission, keep them there as long as possible [33:14]

C2 SLE — Assessment findings:

- Ag-Ab complexes deposit in collagen-rich tissues → many body systems; skin: bruising, rashes, classic butterfly/malar rash (worsened by sun; flares triggered by sun) [46:46]; [47:19]
- Joints (lupus arthritis), kidney (inflammation, renal failure), lungs, heart, eyes, GI tract; not all systems in every patient [48:01]
- Autoantibodies: ANA (anti-nuclear) is a primary one — antibodies vs nuclear/DNA antigens; also autoantibodies vs RBCs and platelets [43:32]; [45:34]
- Labs: ↑ inflammatory markers (CRP, fibrinogen, ESR) = inflammation only; autoantibody list on slide also Anti-DNA, Anti-CCP, Anti-phospholipids, RF [slide 34]; [35:28]
- Dx is a diagnosis of exclusion — 'the great imitator'; no one definitive test; can take years [48:43]

C3 Interventions (autoimmune / SLE):

- Two levels of intervention: stop the underlying autoimmune attack + treat the symptoms [41:39]; [slide 35]
- Stop attack: immunomodulators — immunosuppressive drugs, monoclonal antibodies, corticosteroids; also plasmapheresis [41:58]; [slide 44]
- Treat symptoms/inflammation: NSAIDs (Motrin, ibuprofen, aspirin) — fewer side effects, less powerful; corticosteroids = synthetic cortisol, powerful but many side effects [51:28]

- Use NSAIDs first, add corticosteroids if necessary (NSAIDs won't halt kidney damage) [52:31]
- Keep lupus patients out of the sun — sun exposure triggers exacerbations [47:19]

Flowchart 2 — Hypersensitivity disorders

D1 Nursing problem:

- Problem: immune system reacts to an antigen most people would not react to; not normal/appropriate, usually excessive → can damage host tissue [1:14:17]
- Results in unnecessary inflammation / allergic reactions [class slide 4]
- Hypersensitivity = broad umbrella; allergy (reacts to environmental antigen, e.g., cat dander) and autoimmune disease (reacts to self-antigen) are two specific types [1:14:50]
- Predisposing factors: BOTH genetic predisposition + exposure to triggering antigen (inhaled, ingested, injected, skin contact) [1:16:20]

D2 Type I — allergic rhinitis (hay fever)

- Type I = IgE-mediated; allergic rhinitis = 'hay fever' [1:24:49]
- Findings: runny nose, lots of sneezing, red/itchy/watery eyes [1:24:49]
- Mechanism: mast cells/basophils degranulate in nasal passages / eye mucous membranes [1:27:27]

D3 Type I — asthma

- Type I, IgE-mediated; airway constriction [1:25:12]
- Findings: wheezing, shortness of breath [1:25:12]
- Mechanism: degranulation in the airways [1:27:27]
- Reaction within ~15 min of trigger exposure (e.g., cat-allergic person starts to wheeze in ~15 min) [1:37:56]

D4 Type I — urticaria

- Urticaria = hives; Type I, IgE-mediated [1:25:43]
- Looks like mosquito bites all over the body; very itchy [1:25:43]
- Mechanism: degranulation in the skin [1:27:27]

D5 Type I — atopic dermatitis (eczema)

- Atopic dermatitis = eczema; Type I, IgE-mediated [1:25:12]
- Findings: itchy, scaly, crusty rashes — 'incredibly itchy' [1:25:12]
- Mechanism: degranulation in the skin [1:27:27]

D6 Type I — anaphylaxis

- Most severe Type I disorder; main worry = blood pressure drops → hypotension → can progress to anaphylactic shock; people can die [1:26:14]
- Also bronchoconstriction and some shortness of breath [1:26:14]
- Mechanism: degranulation system-wide → whole-body vasodilation → BP drop [1:27:48]

- Related: angioedema = tongue, lips, part of face swell [1:25:43]
- Textbook, not from her lecture: anaphylaxis emergency care = IM epinephrine first, then airway/oxygen, IV fluids for hypotension, plus antihistamines/corticosteroids

[not covered in lecture — textbook]

not covered in lecture

D7 Type II — blood transfusion reactions

- Type II = antibody-mediated/cytotoxic; uses IgG and IgM (not IgE) [1:30:28]
- Example: type A recipient (has antibodies vs B antigen) given type B blood → recipient antibodies attack the donor cells [1:31:00]
- When antibody binds antigen on donor cell → destroys the donated cell; the binding itself causes the damage [1:31:00]

D8 Type III — SLE (why it is Type III)

- Type III = immune complex-mediated (IgG/IgM); immune complex = antigen-antibody complex [1:32:10]
- Not the binding itself — complexes get deposited in tissue → activate complement → inflammation = the symptoms [1:33:09]
- That is exactly the lupus scenario (ANA binds DNA antigen → complexes deposit in collagen-rich tissue) [44:33]; [1:34:13]

D9 Type IV — contact dermatitis (poison ivy), diabetes mellitus

- Type IV = T-cell-mediated / cellular — only type NOT antibody-mediated; delayed [1:34:44]

- T cells get sensitized to antigen → next exposure, T cells attack → symptoms [1:34:44]
- Examples: poison ivy/poison oak contact dermatitis (rash where touched); type 1 diabetes (sensitized T cells attack pancreatic cells); certain transplant rejections [1:35:16]
- Delayed timing: rash/peak ~1-3 days after exposure (vs Type I/II 15-30 min, Type III ~6 hr) [1:37:56]

D10 Assessment findings (hypersensitivity) — guide says 'Specific to individual disorders'

- Findings are specific to individual disorders; symptoms differ by WHERE mast cells/basophils degranulate (airways→asthma, nose/eyes→rhinitis, skin→eczema/hives, system-wide→anaphylaxis) [1:27:15]
- Diagnostics: allergy testing — skin testing, sticking little bits of antigen under the skin to see which cause a reaction [1:29:22]; [class slide 5]
- All three major immune problems can increase inflammatory markers [class slide 5]
- ★ **Prof emphasis:** on the test she can't ask Type II vs Type IV from symptoms; but anything that sounds like an 'allergy' = Type I IgE-mediated [1:36:33]

D11 Interventions (hypersensitivity):

- Main interventions: get rid of/avoid the trigger + treat the symptoms; we can't desensitize once a person is sensitized [1:39:01]
- Meds for Type I: antihistamines (histamine is a mediator) + anti-inflammatories —

corticosteroids local vs systemic (e.g., fluticasone), others [1:29:05]; [slide 70]

- Help patients identify allergens (skin testing) and avoid them; if unavoidable (common trees, pets) → rely more on medications [1:29:53]
- ★ **Prof emphasis:** we can NEVER stop monitoring patients for allergic reactions to meds — B cells may start making IgE after the 1st, 10th, or 50th exposure (penicillin story)

[1:23:32]